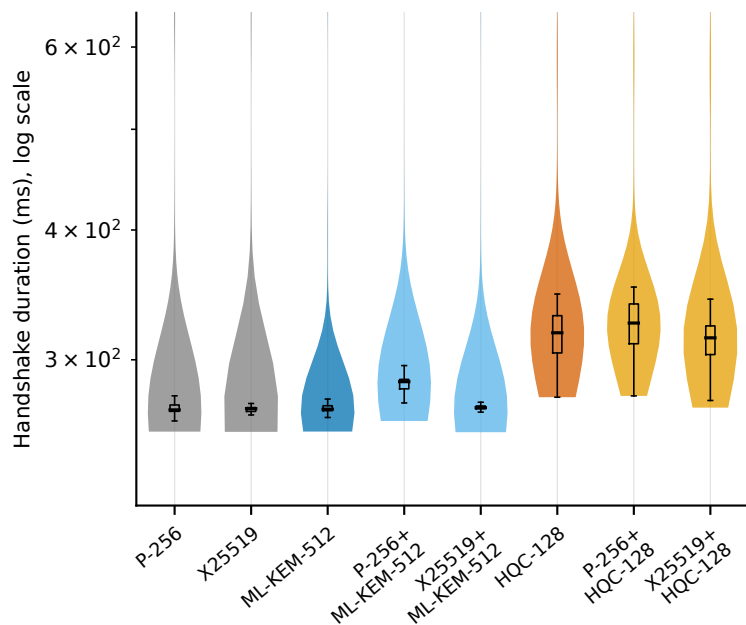
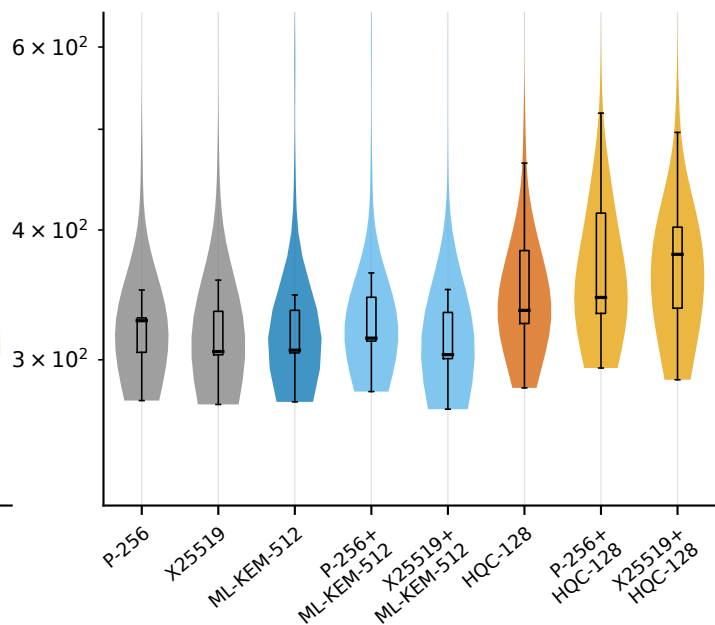


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	295.3	50.7	1.32k	342.4	41.8	1.35k
X25519	291.7	53.7	2.33k	340.1	47.1	1.73k
ML-KEM-512	282.0	35.5	1.31k	331.3	38.9	2.32k
P-256+ML-KEM-512	308.1	46.7	1.41k	347.8	36.4	1.37k
X25519+ML-KEM-512	289.8	44.1	1.32k	332.9	46.0	1.86k
HQC-128	346.7	43.2	1.40k	363.9	34.5	1.42k
P-256+HQC-128	345.6	35.8	1.35k	389.4	39.6	1.53k
X25519+HQC-128	340.1	45.0	1.98k	395.3	41.3	1.54k